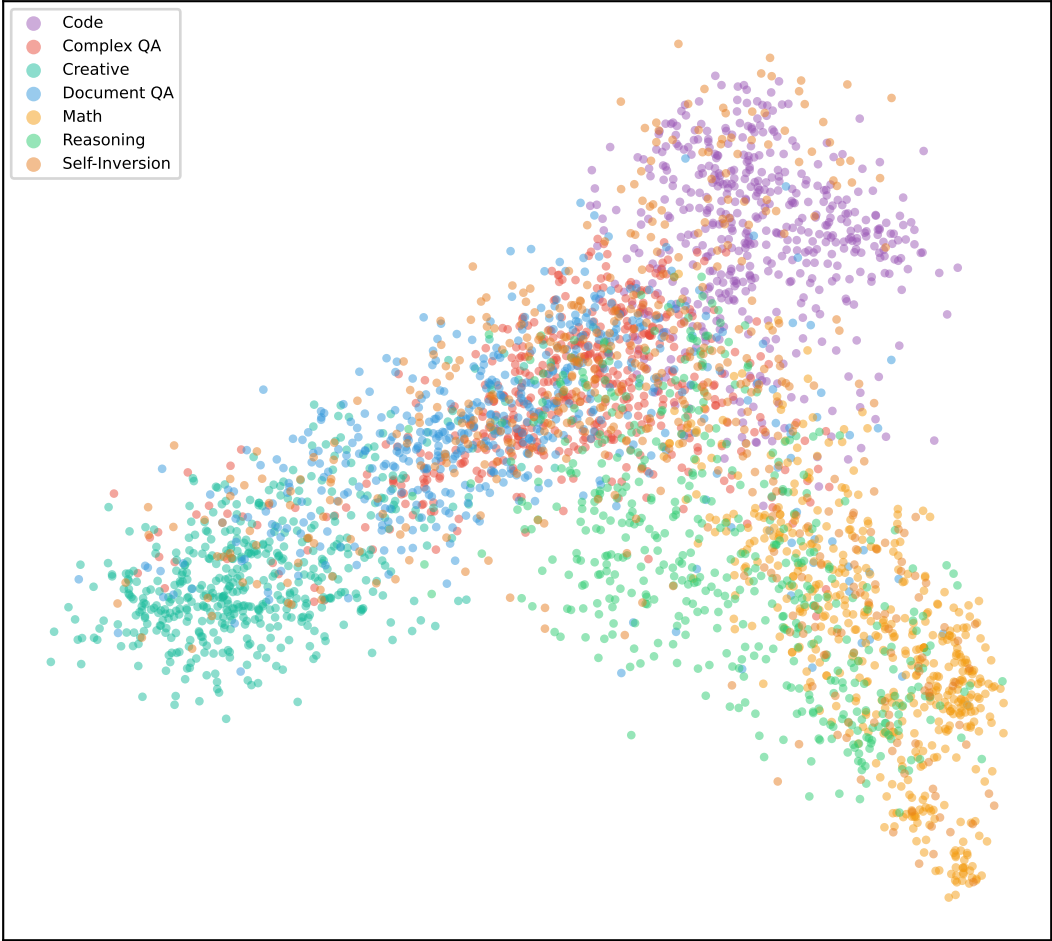
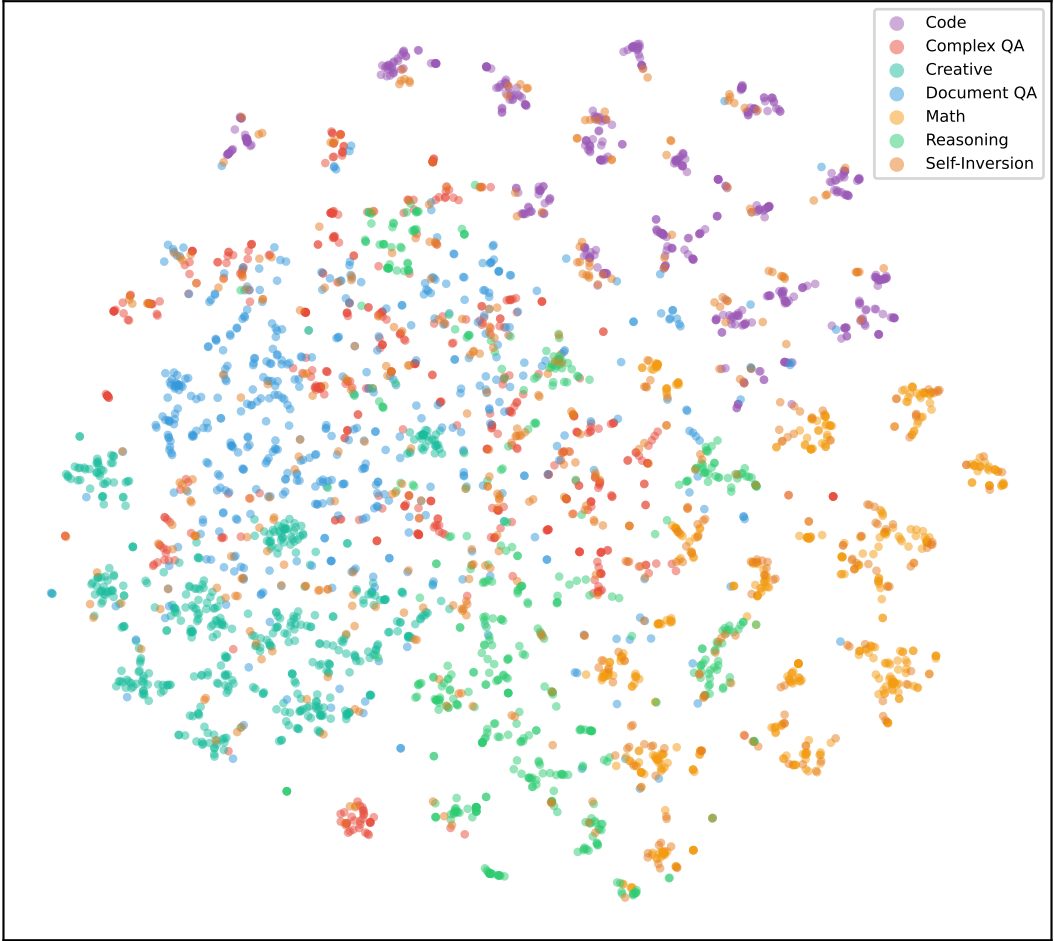


Fig 151: Dimensionality Reduction Method Comparison

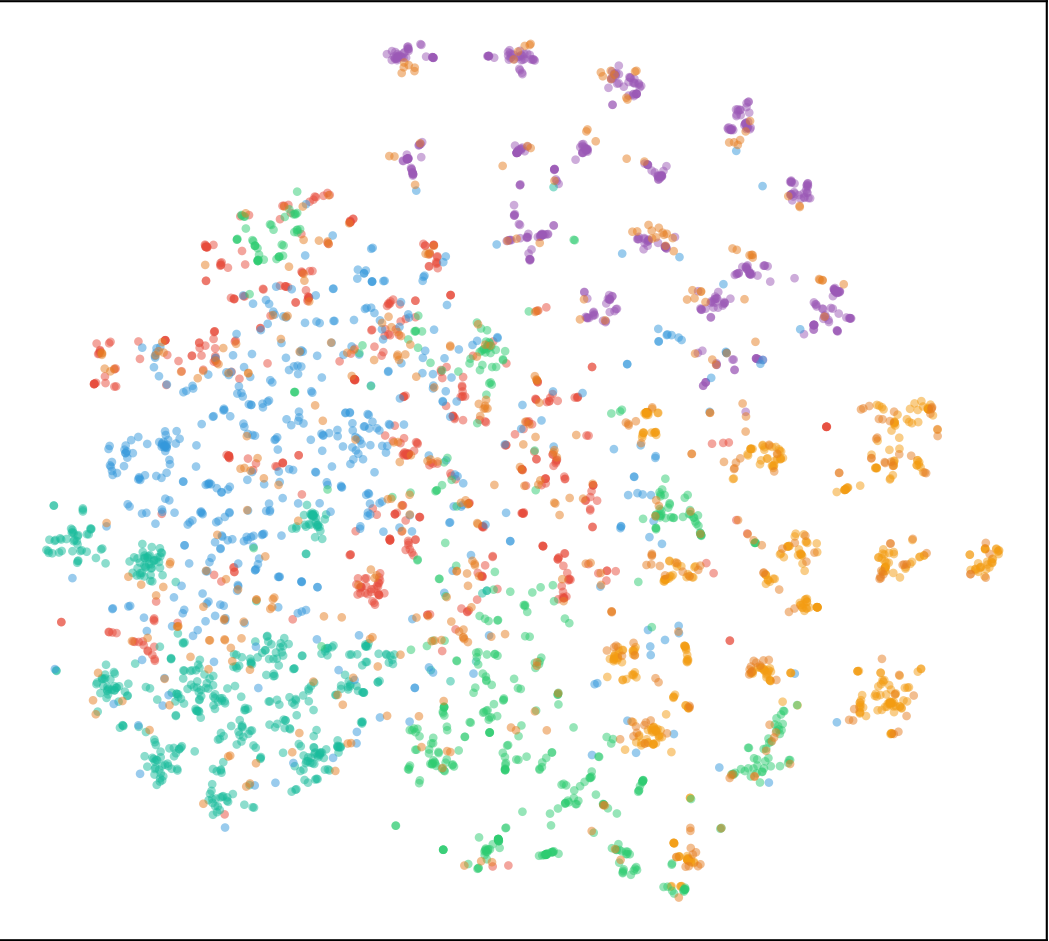
(a) PCA (var=9.2%)



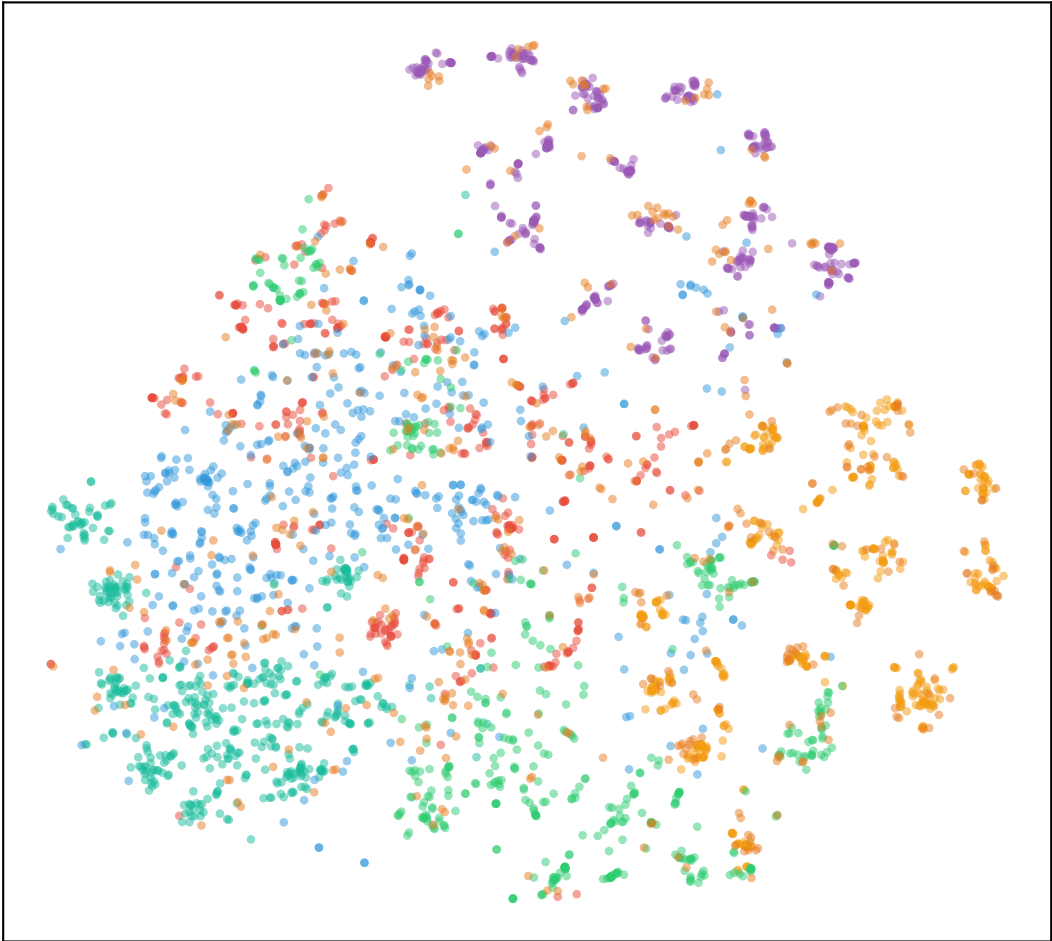
(b) t-SNE (p=15)



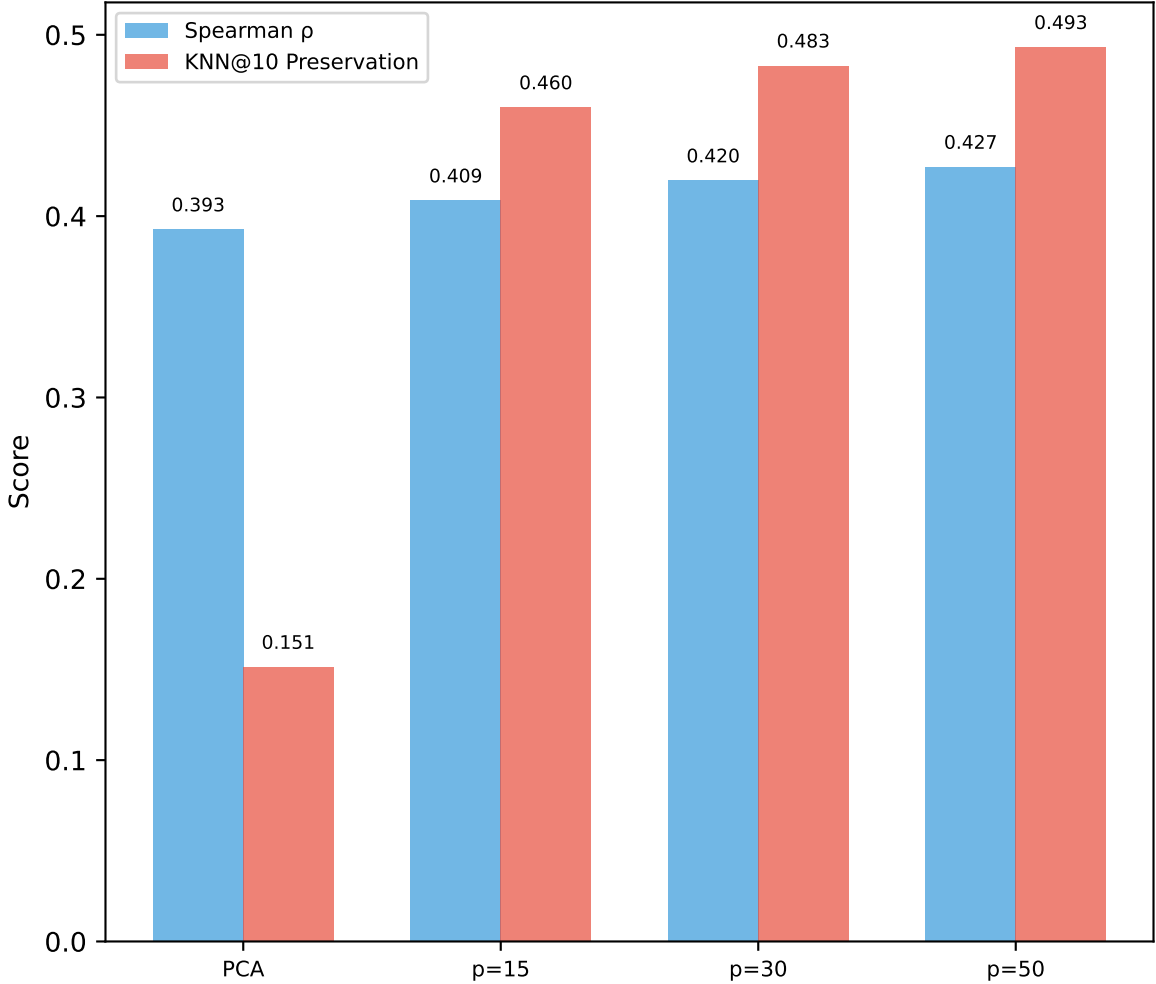
(c) t-SNE (p=30)



(d) t-SNE (p=50)



(e) Reduction Quality Comparison



(f) Summary & Recommendations

Dimensionality Reduction Summary		
Method	Spearman	KNN@10
PCA	0.393	0.151
p=15	0.409	0.460
p=30	0.420	0.483
p=50	0.427	0.493
PCA:		
+ Fast, deterministic		
- Low variance captured (9.2%)		
t-SNE p=30 (default):		
+ Best local structure		
- Non-deterministic, slow		
Recommendation:		
t-SNE p=30 for visualization		
PCA for downstream analysis		